CE, IT, CSE(DS) Department

SEMESTER – IV

Object Oriented Programming (BE04000231)

Mid Semester Exam Syllabus

Contents:

Sr. No.	Content
1	**Basic of Java:** Java Platform and Architecture (JVM, JDK, JRE), Java Program Structure, Compilation, and Execution Java program, Data types (byte, short, int, long, float, double, char, boolean, String, Arrays), variables, keywords, literal, operators, operator precedence and associativity, type conversion
2	**Conditional and looping statements:** simple *if*, *if-else* statement, *else-if* ladder, *switch-case* statement, *for* loop, *while* loop, *do-while* loop, Enhanced for Loop (for-each loop), *break*, *continue*, labelled *break*, labelled *continue*
3	**Basics of Object Oriented Programming:** Data Encapsulation by defining classes, data members, member functions, Data Abstraction, access specifiers, static data member and static member function, defining constructors, objects and references, *this* keyword, returning and passing objects, array of objects, static block, instance block, inner class
4	**Inheritance, Polymorphism and Wrapper classes:** Inheritance, types of inheritance, *super, this* and *final* keywords, method overloading, method overriding, dynamic binding, casting objects, instance of operator, protected data members, String, StringBuffer and StringBuilder class, Wrapper classes for Primitive types, autoboxing/unboxing, Collection framework, Java API Packages, User-defined packages
5	**Interface, Abstract class:** Interface, Partial interface, Multiple inheritance using interface, Abstract class, Abstract methods

Reference Books:

1. **Java: The Complete Reference** – Herbert Schildt, McGraw Hill Education
2. **Head First Java** – Kathy Sierra, Bert Bates, O'Reilly Media.
3. **Core Java: An Integrated Approach** – R. Nageswara Rao, Dreamtech Press
4. **Thinking in Java** – Bruce Eckel, Pearson Education
5. **Object-Oriented Programming with Java** – C. Thomas Wu, Tata McGraw Hill
6. **Programming with Java** – E. Balagurusamy, McGraw Hill Education